

UD 0. Comandos de red en linux

Por cada apartado captura pantalla de que del comando que se ejecuta y de la salida que se ha ejecutado correctamente.

1. Ejercicio: Ver las tarjetas de red físicas en Ubuntu Server

Objetivo: Identificar las tarjetas de red físicas del equipo, diferenciarlas de las interfaces virtuales (lo, docker0, veth, bridges, etc.) y obtener información del driver y del hardware.

Nota: Los nombres de las interfaces físicas suelen ser eth0, enp0s3, ens33, eno1, etc. Las virtuales son lo, docker0, br-*, veth*, virbr0, etc.

Comandos resumen a utilizar

Ver todas las interfaces	<code>ip -br link show</code>
Ver solo hardware PCI de red	<code>lspci grep -i ethernet</code>
Ver hardware de red resumido	<code>sudo lshw -class network -short</code>
Ver detección del kernel	<code>dmesg grep -i eth</code>
Ver driver cargado	<code>lsmod grep -i <driver></code>
Ver driver de una interfaz	<code>sudo ethtool -i <interfaz></code>

1.1. Listar todas las interfaces de red (físicas y virtuales).

```
alumno@servidor:~$ ip -br link show
lo                UNKNOWN          00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
enp0s3            UP                08:00:27:a6:fe:cc <BROADCAST,MULTICAST,UP,LOWER_UP>
enp0s8            UP                08:00:27:3a:69:61 <BROADCAST,MULTICAST,UP,LOWER_UP>
alumno@servidor:~$
```

1.2. Ver solo las interfaces que tienen un dispositivo PCI físico asociado

```
alumno@servidor:~$ sudo dmesg | grep PCI
[ 0.017026] [mem 0x80000000-0xfefbffff] available for PCI devices
[ 0.223152] acpihp: ACPI Hot Plug PCI Controller Driver version: 0.5
[ 0.223601] PCI: Using configuration type 1 for base access
[ 0.263167] PCI: Using host bridge windows from ACPI; if necessary, use "pci=nocrs" and report a bug
[ 0.263766] PCI: Using E820 reservations for host bridge windows
[ 0.271681] ACPI: PCI Root Bridge [PCI0] (domain 0000 [bus 00-ff])
[ 0.275685] PCI host bridge to bus 0000:00
[ 0.278780] pci 0000:00:00.0: [8086:1237] type 00 class 0x060000 conventional PCI endpoint
[ 0.279389] pci 0000:00:01.0: [8086:7000] type 00 class 0x060100 conventional PCI endpoint
[ 0.280916] pci 0000:00:01.1: [8086:7111] type 00 class 0x01018a conventional PCI endpoint
[ 0.284561] pci 0000:00:02.0: [15ad:0405] type 00 class 0x030000 conventional PCI endpoint
[ 0.294390] pci 0000:00:03.0: [8086:100e] type 00 class 0x020000 conventional PCI endpoint
[ 0.300796] pci 0000:00:04.0: [80ee:cafe] type 00 class 0x088000 conventional PCI endpoint
[ 0.308439] pci 0000:00:06.0: [106b:003f] type 00 class 0x0c0310 conventional PCI endpoint
[ 0.315232] pci 0000:00:07.0: [8086:7113] type 00 class 0x068000 conventional PCI endpoint
[ 0.318514] pci 0000:00:08.0: [8086:100e] type 00 class 0x020000 conventional PCI endpoint
[ 0.327048] pci 0000:00:0d.0: [8086:2829] type 00 class 0x010601 conventional PCI endpoint
[ 0.347326] ACPI: PCI: Interrupt link LNKA configured for IRQ 11
[ 0.348047] ACPI: PCI: Interrupt link LNKB configured for IRQ 10
[ 0.349243] ACPI: PCI: Interrupt link LNKC configured for IRQ 9
[ 0.349243] ACPI: PCI: Interrupt link LNKD configured for IRQ 11
[ 0.359489] PCI: Using ACPI for IRQ routing
[ 0.359820] PCI: pci cache line size set to 64 bytes
[ 0.396614] pci 0000:00:00.0: Limiting direct PCI/PCI transfers
[ 0.411776] PCI: CLS 0 bytes, default 64
[ 0.464935] shpchp: Standard Hot Plug PCI Controller Driver version: 0.4
[ 0.480837] ohci-pci 0000:00:06.0: OHCI PCI host controller
[ 0.559050] usb usb1: Product: OHCI PCI host controller
[ 1.239727] e1000 0000:00:03.0 eth0: (PCI:33MHz:32-bit) 08:00:27:a6:fe:cc
[ 1.602520] e1000 0000:00:08.0 eth1: (PCI:33MHz:32-bit) 08:00:27:3a:69:61
```

1.3. Obtener información detallada del hardware de red

```
alumno@servidor:~$ lspci | grep Ethernet
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 02)
00:08.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 02)
```

1.4. Ver los mensajes del kernel al detectar las tarjetas

```
alumno@servidor:~$ sudo dmesg | grep eth
[ 1.239727] e1000 0000:00:03.0 eth0: (PCI:33MHz:32-bit) 08:00:27:a6:fe:cc
[ 1.240144] e1000 0000:00:03.0 eth0: Intel(R) PRO/1000 Network Connection
[ 1.602520] e1000 0000:00:08.0 eth1: (PCI:33MHz:32-bit) 08:00:27:3a:69:61
[ 1.602872] e1000 0000:00:08.0 eth1: Intel(R) PRO/1000 Network Connection
[ 1.604304] e1000 0000:00:03.0 enp0s3: renamed from eth0
[ 1.612952] e1000 0000:00:08.0 enp0s8: renamed from eth1
```

1.5. Comprobar los módulos del kernel cargados para las tarjetas

```
alumno@servidor:~$ lsmod
Module                  Size      Used by
tls                     155648    0
intel_rapl_msrm         20480     0
intel_rapl_common       40960     1 intel_rapl_msrm
intel_uncore_frequency  16384     0
intel_pmc_core          118784    0
intel_vsec              20480     1 intel_pmc_core
binfmt_misc            24576     1
pmt_telemetry           16384     1 intel_pmc_core
pmt_class               12288     1 pmt_telemetry
kvm_intel               487424    0
kvm                     1404928   1 kvm_intel
vmwgfx                  438272    0
irqbypass              12288     1 kvm
i2c_piix4               32768     0
drm_ttm_helper          12288     1 vmwgfx
vboxguest               57344     0
ttm                     110592    2 vmwgfx,drm_ttm_helper
joydev                  32768     0
input_leds              12288     0
mac_hid                 12288     0
serio_raw               20480     0
cfg80211                1323008   0
sch_fq_codel            24576     3
dm_multipath            45056     0
efi_pstore              12288     0
nfnetlink               20480     2
dmi_sysfs               24576     0
ip_tables               32768     0
x_tables                65536     1 ip_tables
autofs4                 57344     2
btrfs                  2015232   0
blake2b_generic         24576     0
raid10                  73728     0
raid456                 192512    0
async_raid6_recov       20480     1 raid456
async_memcpy            16384     2 raid456,async_raid6_recov
async_pq                20480     2 raid456,async_raid6_recov
async_xor               16384     3 async_pq,raid456,async_raid6_recov
async_tx               16384     5 async_pq,async_memcpy,async_xor,raid456,async_raid6_recov
xor                    20480     2 async_xor,btrfs
raid6_pq               126976    4 async_pq,btrfs,raid456,async_raid6_recov
libcrc32c               12288     2 btrfs,raid456
raid1                   57344     0
raid0                   24576     0
hid_generic             12288     0
usbhid                  77824     0
hid                     180224    2 usbhid,hid_generic
crct10dif_pclmul        12288     1
crc32_pclmul            12288     0
polyval_clmulni         12288     0
polyval_generic         12288     1 polyval_clmulni
ghash_clmulni_intel     16384     0
sha256_ssse3            32768     0
sha1_ssse3              32768     0
psmouse                217088    0
ahci                     49152     2
e1000                   180224    0
libahci                 53248     1 ahci
video                   73728     0
wmi                     28672     1 video
pata_acpi               12288     0
aesni_intel             356352    0
crypto_simd             16384     1 aesni_intel
cryptd                  24576     2 crypto_simd,ghash_clmulni_intel
```

1.6. Ver el driver y detalles de una interfaz concreta

```
alumno@servidor:~$ sudo ethtool -i enp0s8
driver: e1000
version: 6.8.0-47-generic
firmware-version:
expansion-rom-version:
bus-info: 0000:00:08.0
supports-statistics: yes
supports-test: yes
supports-eprom-access: yes
supports-register-dump: yes
supports-priv-flags: no
```

Responde a estas preguntas:

¿Cuántas tarjetas de red físicas tiene el equipo?

```
alumno@servidor:~$ sudo dmesg | grep eth
[ 1.239727] e1000 0000:00:03.0 eth0: (PCI:33MHz:32-bit) 08:00:27:a6:fe:cc
[ 1.240144] e1000 0000:00:03.0 eth0: Intel(R) PRO/1000 Network Connection
[ 1.602520] e1000 0000:00:08.0 eth1: (PCI:33MHz:32-bit) 08:00:27:3a:69:61
[ 1.602872] e1000 0000:00:08.0 eth1: Intel(R) PRO/1000 Network Connection
[ 1.604304] e1000 0000:00:03.0 enp0s3: renamed from eth0
[ 1.612952] e1000 0000:00:08.0 enp0s8: renamed from eth1
```

Tiene 6

¿Cuál es la dirección MAC de cada una?

```
alumno@servidor:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:a6:fe:cc brd ff:ff:ff:ff:ff:ff
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:3a:69:61 brd ff:ff:ff:ff:ff:ff
```

La MAC se encuentran después de la palabra link/ether

¿Qué driver utiliza cada tarjeta?

```
alumno@servidor:~$ sudo dmesg | grep eth
[ 1.239727] e1000 0000:00:03.0 eth0: (PCI:33MHz:32-bit) 08:00:27:a6:fe:cc
[ 1.240144] e1000 0000:00:03.0 eth0: Intel(R) PRO/1000 Network Connection
[ 1.602520] e1000 0000:00:08.0 eth1: (PCI:33MHz:32-bit) 08:00:27:3a:69:61
[ 1.602872] e1000 0000:00:08.0 eth1: Intel(R) PRO/1000 Network Connection
[ 1.604304] e1000 0000:00:03.0 enp0s3: renamed from eth0
[ 1.612952] e1000 0000:00:08.0 enp0s8: renamed from eth1
```

El driver e1000

¿Qué interfaces son virtuales y por qué?

```
alumno@servidor:~$ sudo dmesg | grep eth
[ 1.239727] e1000 0000:00:03.0 eth0: (PCI:33MHz:32-bit) 08:00:27:a6:fe:cc
[ 1.240144] e1000 0000:00:03.0 eth0: Intel(R) PRO/1000 Network Connection
[ 1.602520] e1000 0000:00:08.0 eth1: (PCI:33MHz:32-bit) 08:00:27:3a:69:61
[ 1.602872] e1000 0000:00:08.0 eth1: Intel(R) PRO/1000 Network Connection
[ 1.604304] e1000 0000:00:03.0 enp0s3: renamed from eth0
[ 1.612952] e1000 0000:00:08.0 enp0s8: renamed from eth1
```

Solo la interfaz loopback, porque ya que no está asociada a ningún componente de hardware físico

2. Ejercicio : Ver todas las interfaces de red disponibles

Resumen de comandos a utilizar

Ver interfaces	ip link show / ip addr show
Ver resumen	ip -br link / ip -br addr
Activar interfaz	sudo ip link set <if> up
Desactivar interfaz	sudo ip link set <if> down
Añadir IP	sudo ip addr add <IP>/<prefijo> dev <if>
Eliminar IP	sudo ip addr del <IP>/<prefijo> dev <if>
Ver tabla ARP	ip neigh show

2.1. Muestra todas las interfaces con su estado:

```
davidrodera@PC-DAROVA:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
3: tailscale0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1280 qdisc fq_codel state UNKNOWN mode DEFAULT group default qlen 500
    link/none
```

2.2. Muestra solo las interfaces activas (UP):

```
davidrodera@PC-DAROVA:~$ ip link show up
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
3: tailscale0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1280 qdisc fq_codel state UNKNOWN mode DEFAULT group default qlen 500
    link/none
```

2.3. Muestra las direcciones IP de todas las interfaces:

```
davidrodera@PC-DAROVA:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.133/24 brd 192.168.1.255 scope global dynamic noprefixroute enp3s0
        valid_lft 86086sec preferred_lft 86086sec
3: tailscale0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1280 qdisc fq_codel state UNKNOWN group default qlen 500
    link/none
    inet 100.126.6.126/32 scope global tailscale0
        valid_lft forever preferred_lft forever
    inet6 fd7a:115c:a1e0:ac33:67e/128 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::adc4:b4f7:2b06:19cd/64 scope link stable-privacy
        valid_lft forever preferred_lft forever
```

3. Ejercicio : Desactivar una tarjeta de red

3.1. Comprueba que la interfaz está activa (ej. enp3s0):

```
davidrodera@PC-DAROVA:~$ ip link show enp3s0
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
```

3.2. Desactívala:

```
davidrodera@PC-DAROVA:~$ sudo ip link set enp3s0 down
[sudo] contraseña para davidrodera:
```

3.3. Verifica que ahora está DOWN:

```
davidrodera@PC-DAROVA:~$ ip link show up
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
3: tailscale0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1280 qdisc fq_codel state UNKNOWN mode DEFAULT group default qlen 500
    link/none
```

No sale la interfaz dentro de las activas.

4. Ejercicio : Activar una interfaz de red

Objetivo: Poner en estado UP una interfaz que está DOWN.

4.1. Comprueba el estado actual de la interfaz (supongamos enp3s0):

```
davidrodera@PC-DAROVA:~$ ip link show enp3s0
2: enp3s0: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
```

4.2. Actívala:

```
davidrodera@PC-DAROVA:~$ sudo ip link set enp3s0 up
```

4.3. Verifica que ahora está UP:

```
davidrodera@PC-DAROVA:~$ ip link show enp3s0
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
```

5. Configurar varias direcciones IP en una misma interfaz

Objetivo: Asignar dos direcciones IP adicionales a una interfaz (sin eliminar la que ya tiene).

5.1. Asegúrate de que la interfaz está activa (ej. enp3s0):

```
davidrodera@PC-DAROVA:~$ ip link show enp3s0
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
```

5.2. Añade una primera IP:

```
davidrodera@PC-DAROVA:~$ sudo ip addr add 192.168.1.134/24 dev enp3s0
davidrodera@PC-DAROVA:~$ |
```

5.3. Añade una segunda IP:

```
davidrodera@PC-DAROVA:~$ sudo ip addr add 192.168.1.135/24 dev enp3s0
davidrodera@PC-DAROVA:~$ |
```

5.4. Verifica que ambas están asignadas:

```
davidrodera@PC-DAROVA:~$ ip addr show enp3s0
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.133/24 brd 192.168.1.255 scope global dynamic noprefixroute enp3s0
        valid_lft 85952sec preferred_lft 85952sec
    inet 192.168.1.134/24 scope global secondary enp3s0
        valid_lft forever preferred_lft forever
    inet 192.168.1.135/24 scope global secondary enp3s0
        valid_lft forever preferred_lft forever
```

6. Ejercicio : Eliminar una dirección IP de una interfaz

Objetivo: Borrar una de las direcciones IP previamente asignadas.

6.1. Muestra las IPs actuales de la interfaz (ej. enp3s0):

```
davidrodera@PC-DAROVA:~$ ip addr show enp3s0
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.133/24 brd 192.168.1.255 scope global dynamic noprefixroute enp3s0
        valid_lft 85952sec preferred_lft 85952sec
    inet 192.168.1.134/24 scope global secondary enp3s0
        valid_lft forever preferred_lft forever
    inet 192.168.1.135/24 scope global secondary enp3s0
        valid_lft forever preferred_lft forever
```


6.2. Elimina una IP

```
davidrodera@PC-DAROVA:~$ sudo ip addr del 192.168.1.135/24 dev enp3s0
```

6.3. Verifica que ya no está:

```
davidrodera@PC-DAROVA:~$ ip addr show enp3s0
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 18:c0:4d:ba:17:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.133/24 brd 192.168.1.255 scope global dynamic noprefixroute enp3s0
        valid_lft 85872sec preferred_lft 85872sec
    inet 192.168.1.134/24 scope global secondary enp3s0
        valid_lft forever preferred_lft forever
```

7. Ejercicio 6: Ver la tabla ARP (vecinos)

Objetivo: Visualizar las entradas ARP del sistema, tanto las aprendidas dinámicamente como las estáticas.

7.1. Muestra la tabla ARP actual:

```
davidrodera@PC-DAROVA:~$ ip neigh show
192.168.1.1 dev enp3s0 lladdr 54:ce:82:90:d1:74 REACHABLE
```